



TUBE CLEANING

CHECKLIST (Automated)



- ☐ All team members must understand and practice hand signals for “Turn on Water”, “Pressure Up/OK to begin blasting”, and “Cut Pressure”

Indexer / Positioner Setup

- ☐ Proper mechanical attachment to tubesheet/channel head/scaffold
- ☐ For multi-lance operations, demonstrate matching guide tubes with exchanger pitch
- ☐ Installation of guide tube assembly within 1” or less of tubesheet
- ☐ Demonstrates ability to match proper sized AWD collet with lance

Lance Drive Installation

- ☐ Demonstrate proper installation of lance drive to indexer/positioner
- ☐ Demonstrate proper procedure for loading lance(s)
- ☐ Demonstrate proper installation of AWD collet
- ☐ Proper connection of control hoses (routed to minimize trip hazards and contamination with hazardous materials)

Control Panel Operations

- ☐ Positions controls away from high pressure waterjet and hazardous material from exchanger, maintaining clear line of sight to guide tube(s)
- ☐ Proper connection and directional confirmation of control hoses / panel
- ☐ Proper connection and ease of access of dump valve



TUBE CLEANING

CHECKLIST (Automated) – CONT.



Control Panel Operation (cont.)

- ☐ Proper installation of lance(s) to dump valve
- ☐ Proper operation of indexing functions

Initiates Pressure Up

- ☐ Demonstrative in use of proper hand signals
- ☐ Demonstrates proper installation of hose stops to prevent nozzle exiting the far end of the tube
- ☐ Keeps dump valve activated until proper pressure is attained
- ☐ Tests dump valve before starting to work
- ☐ Demonstrates proper operation including speed adjustment
- ☐ Demonstrates ability to properly cycle the lances through multiple tubes

System Pressure Down

- ☐ Demonstrative in use of proper hand signal
- ☐ Maintains control of equipment until pump is shut down